Breast Cancer Classification and Segmentation Using Deep Learning on Ultrasound Images: A Leakage-Free Benchmark, an Improved Segmenter, and a Rigorously Evaluated Segmentation-Guided Classifier

COMP9444 Group AreuOk? Project 047

Abstract

Breast ultrasound (BUS) is a safe, low-cost imaging modality for breast-cancer screening, but it is difficult to interpret and its main public benchmark, BUSI, is small, imbalanced and contains many near-duplicate images. We study lesion *segmentation* and image *classification* on BUSI under a single, carefully cleaned, *leakage-free* evaluation protocol. Using perceptual hashing we find that roughly one third of BUSI images belong to near-duplicate clusters and that an exact cross-class duplicate exists; we therefore split the data at the cluster level. Under this protocol we reproduce thirteen published models and add two of our own: an improved pretrained-encoder U-Net and an improved segmentation-guided (dual-stage) classifier. A controlled ablation shows that an ImageNet-pretrained encoder alone raises segmentation Dice from 0.648 to 0.765 and is responsible for essentially all of our segmenter's gain, whereas a Focal-Tversky loss, test-time augmentation and attention gates do not reliably improve overlap. A controlled Stage-1 substitution shows the dual-stage pipeline is limited by its *classifier*, not its segmenter. When candidate classifier improvements are evaluated rigorously (out-of-fold training masks, five-seed ensembling and bootstrap 95% confidence intervals), hand-crafted shape/radiomics features give no reliable gain and only ensembling survives, reaching macro-F1 0.875 (95% CI $[0.80, 0.94]$). Our reproductions of published BUSI models score below their reported numbers, which we attribute to removing near-duplicate leakage. This is an experimental decision-support study, not a system for autonomous clinical diagnosis.

1 Introduction

Breast cancer is one of the leading causes of cancer-related death among women worldwide, and early detection remains the single most effective way to reduce mortality and improve outcomes [18, 19]. Among the common imaging modalities, breast ultrasound (BUS) is safe, inexpensive and radiation-free, and is particularly valuable for younger patients and for dense breast tissue, where the sensitivity of mammography is reduced.

Despite these advantages, BUS images are hard to read. They suffer from *speckle noise*, *low contrast*, *acoustic shadowing* and *blurred lesion boundaries*, and the appearance of a lesion (its shape, echogenicity and texture) varies greatly between and within the benign and malignant categories [1, 18]. As a result, manual assessment is slow and operator-dependent, with appreciable inter-observer variability. Several BUSI images additionally contain *burned-in annotations*, calipers and text (for example "RT LOQ" or "RIGHT BREAST"), which a model could exploit as a spurious shortcut. These difficulties motivate computer-aided diagnosis (CAD) systems that can support, rather than replace, the radiologist.

Classification and segmentation are complementary views of the same clinical problem. *Classification* answers *what* a scan shows – normal tissue, a benign mass, or a malignant mass – while *segmentation* answers *where* the lesion is, delineating its boundary at the pixel level. A lesion mask supports interpretability and downstream measurement, and it can be used to focus a classifier on the lesion rather than on surrounding clutter, which links the two tasks in a single pipeline.

Problem statement. Following the Project 047 brief, we build and evaluate a deep-learning system on the Breast Ultrasound Images (BUSI) dataset [1] that addresses both tasks: three-class classification (`normal`/`benign`/`malignant`) and binary lesion segmentation. Our central concern is *trustworthy comparison*: BUSI is small, imbalanced and, as we show, riddled with near-duplicate images, so a careless evaluation can substantially inflate reported scores. We therefore build a single cleaned, leakage-free evaluation protocol and compare every model under it.

Research questions. The study is organised around four questions.

- **RQ1.** How do established segmentation and classification architectures perform under a common, leakage-free BUSI evaluation protocol, and how do those numbers compare with the values reported in the original papers?
- **RQ2.** Which design decisions most strongly affect segmentation performance on the small BUSI dataset (transfer learning, loss function, attention, inference-time processing)?
- **RQ3.** Does improving lesion segmentation reliably improve downstream segmentation-guided classification?
- **RQ4.** Are apparent improvements stable across random seeds and statistical resampling, or are they artefacts of a single training run on a small test set?

Contributions. Grounded in the actual work, our contributions are: (i) a near-duplicate analysis of BUSI (perceptual hashing and cluster detection) and a *cluster-grouped, class-stratified* leakage-free split; (ii) a unified reproduction of thirteen published models – eight segmentation architectures, three classification backbones, one dual-stage pipeline and one multi-task network – all trained and evaluated under this one protocol; (iii) an improved pretrained-encoder segmenter with a *controlled ablation* that isolates the contribution of each ingredient; (iv) an improved dual-stage (segmentation-guided) classifier, including a controlled Stage-1 substitution and a ground-truth-ROI upper-bound diagnostic; and (v) a *rigorous evaluation protocol* for the dual-stage system that uses out-of-fold training-mask generation, multi-seed ensembling and bootstrap 95% confidence intervals, which we use to separate real improvements from noise.

Report organisation. Section 2 reviews related work and identifies the gaps this project targets. Section 3 describes the dataset and our exploratory data analysis (EDA), and maps each observation to a modelling decision. Section 4 details the models, our own components and the loss functions. Section 5 specifies the experimental setup: splits, hyper-parameters, metrics and the statistical protocol. Section 6 reports results and error analysis, Section 7 discusses their implications and limitations, and Section 8 concludes.

2 Literature Review

We organise the literature thematically and then synthesise the gaps that motivate our methodology.

2.1 Breast-ultrasound CAD and the BUSI dataset

Classical CAD for breast ultrasound followed a pipeline of pre-processing, segmentation, feature extraction and classification; the surveys of Cheng et al. [19] and Xian et al. [18] review two decades of such methods and recur to the same difficulties – speckle noise, low signal-to-noise ratio, low contrast and fuzzy boundaries – that still dominate today. The public BUSI dataset of Al-Dhabyani et al. [1] has become the de-facto benchmark, providing `normal`/`benign`/`malignant` images with pixel-level masks and thus supporting both tasks. Because BUSI is small, augmentation matters: the same authors showed that traditional and GAN-based augmentation improve classification on breast ultrasound [24]. Crucially, a documented data-quality issue – repeated and near-duplicate frames, some with conflicting labels – has been raised for BUSI [28]; we quantify this directly in Section 3.5 and design our evaluation around it.

2.2 CNN-based classification

Modern image classification is dominated by deep convolutional backbones. ResNet [4] introduced residual (identity-shortcut) connections that make very deep networks trainable; DenseNet [5] connects each layer to all subsequent layers, promoting feature reuse and parameter efficiency, which is attractive for small datasets; and EfficientNet [6] uses compound scaling of depth, width and resolution to reach high accuracy at low cost, with its smaller variants (B0/B1) transferring well to medical data. These backbones have been applied directly to BUSI: Latha et al. [14] fine-tuned EfficientNet-B7 with heavy pre-processing and Grad-CAM [23] explainability and reported very high accuracy (around 99%); such near-perfect numbers on so small a dataset must be treated cautiously because of possible data leakage, an observation that directly motivates our protocol.

2.3 U-Net-based segmentation

The U-Net [3] encoder–decoder with skip connections is the foundational biomedical-segmentation architecture and trains well from few annotated images. Refinements include UNet++ [8], which adds nested dense skip pathways with deep supervision to bridge the encoder–decoder semantic gap, and UNet 3+ [10], which fuses features at full scale. Yap et al. [2] were among the first to apply deep learning to BUS lesion detection, comparing a patch-based LeNet, a U-Net, and a transfer-learned fully convolutional AlexNet (FCN-AlexNet), with the FCN performing best.

2.4 Attention and multi-scale segmentation

Attention U-Net [9] inserts additive attention gates on the skip connections so the network learns *where to look*, suppressing irrelevant background. From semantic segmentation, DeepLabV3+ [7] combines atrous (dilated) separable convolutions and Atrous Spatial Pyramid Pooling (ASPP) for multi-scale context with a light decoder. TransUNet [11] couples a Transformer encoder with a U-Net decoder to model long-range dependencies, but Transformers are typically data-hungry and require large-scale pre-training. Two architectures are tailored to BUS: AAU-Net [12] replaces standard convolutions with a hybrid adaptive attention module for blurred boundaries and irregular malignant shapes, and CResU-Net [13] schedules specialised residual/dense encoder blocks and reports the strongest BUSI Dice in this literature.

2.5 Transfer learning and self-configuring pipelines

Transfer learning from ImageNet is a standard remedy for small medical datasets, reusing general low-level features and stabilising optimisation. nnU-Net [20] is a self-configuring U-Net pipeline that automatically selects pre-processing, architecture and training and remains a strong medical-segmentation baseline. The Segment Anything Model (SAM) [21] and its medical adaptation Med-SAM [22] are promptable foundation segmenters with strong zero-shot transfer; they are trained mostly on natural or general medical images and typically require prompts or adaptation, so we treat them as context rather than as baselines.

2.6 Joint and dual-stage segmentation–classification

Most relevant to our combined system are methods that address both tasks. Bruno et al. [17] propose a *sequential* dual-stage framework: a segmentation network localises the lesion, a region of interest (ROI) is cropped from the predicted mask, and a compact CNN then classifies the ROI as benign or malignant. Lu et al. [25] instead use *joint* multi-task learning with a shared encoder and separate segmentation (object-contextual-attention) and classification heads. These two lines – sequential and joint – frame the design space we explore.

2.7 Evaluation weaknesses in small medical-imaging datasets

A recurring methodological weakness underlies much of this literature. Reported BUSI scores are often obtained with different splits and pre-processing, preventing fair comparison; many studies report a *single* training run without seeds or confidence intervals; and near-duplicate images can leak between train and test under a naive split, inflating scores. On a test set as small as BUSI's, single-run differences of a few points may not be statistically resolved. These weaknesses are precisely what our protocol addresses.

2.8 Research gaps and project motivation

Synthesising the above, six gaps motivate this project and connect directly to our methodology. First, **BUSI is small and imbalanced**, so results are sensitive to overfitting and to the choice of metric; we therefore use transfer learning, class-weighted losses and macro-averaged metrics. Second, **near-duplicate images cause train/test leakage**; we detect duplicates and split at the cluster level (Section 3.5). Third, **published models use different splits and pre-processing**, preventing fair comparison; we put every model through one shared pipeline. Fourth, **many studies report a single run**; we report multi-seed statistics for our final pipeline. Fifth, **confidence intervals are usually absent**; we add bootstrap 95% CIs. Sixth, **it is unclear whether better segmentation implies better downstream classification**; we test this directly with a controlled Stage-1 substitution and a ground-truth-ROI upper bound. Finally, because high published BUSI scores may not survive a leakage-free split, we expect – and report – our reproductions to be lower but more trustworthy.

3 Dataset and Exploratory Data Analysis

All EDA numbers in this section are computed in `COMP9444_EDA.ipynb` (Sections 2–3) from a cached per-image metadata table; we quote them here and connect each observation to a modelling decision.

3.1 Dataset composition

BUSI [1] was collected in 2018 at Baheya Hospital, Cairo, from 600 female patients aged 25–75. It contains 780 grayscale ultrasound scans stored as PNG (average size $\approx 500 \times 500$ px), each with an image-level label in {`normal`, `benign`, `malignant`} and one or more pixel-level ground-truth lesion

Table 1: Comparative summary of the most relevant prior work. "Reported result" lists a paper-stated BUSI number where available to us; "n/r" = not reported (to us). Values are as stated by the cited works, not re-measured here.

Ref	Yr	Task	Key architecture / idea	Reported (BUSI)	Limitation for this project
BUSI [1]	2020	Both	Dataset + masks (n/a)	n/a	Small; near-duplicates; imbalance
Yap [2]	2018	Seg/Det	Patch-LeNet, U-Net, FCN-AlexNet	n/r	Detection focus; not BUSI; older nets
U-Net [3]	2015	Seg	Encoder–decoder + skip connections	n/r	Not BUS-specific; no speckle handling
DeepLabV3+ [7]	2018	Seg	ASPP + atrous separable conv	n/r	Natural-image net; tuning for BUS
Attn. U-Net [9]	2018	Seg	Additive attention gates on skips	n/r	Designed for CT; adds complexity
AAU-Net [12]	2023	Seg	Hybrid adaptive attention (BUS)	Dice 0.775	Complex; costly to reproduce faithfully
CResU-Net [13]	2024	Seg	Co-Block / residual block schedule (BUS)	Dice 0.829	Reported CV without de-duplication
nnU-Net [20]	2021	Seg	Self-configuring U-Net pipeline	n/r	Compute-heavy; general (not BUS)
ResNet [4]	2016	Cls	Residual shortcuts	n/r	Natural-image backbone; overfits small data
DenseNet [5]	2017	Cls	Dense connectivity / feature reuse	n/r	As above; needs regularisation
EfficientNet [6]	2019	Cls	Compound scaling (MBConv)	n/r	High-capacity variants overfit BUSI
Bruno [17]	2025	Both	Dual-stage seg → ROI → cls	Dice 0.769; F1 0.944	2-class; pipeline more complex
MTL-OCA [25]	2025	Both	Shared encoder + OCA + cls head	n/r	Details/numbers to confirm; shares capacity

masks; `normal` scans carry an all-black mask. Table 2 summarises the dataset. Every image has at least one mask (no orphan files), and 647 images contain a real (non-empty) lesion.

A small number of images ship with *multiple* mask files (17 images, up to three masks each, mostly benign). For binary segmentation we merge these into a single target by pixel-wise *union* (`eda_utils.load_union_mask`). We use the data in three ways: **three-class classification** uses all classes; **lesion segmentation** uses only the benign+malignant images that contain a lesion; and the **two-class dual-stage** classifier operates on the benign/malignant lesion images. The exact split counts are given in Section 5.

3.2 Class imbalance

The classes are imbalanced (Fig. 1): `benign` 437 (56.0%), `malignant` 210 (26.9%) and `normal` 133 (17.1%); benign is roughly three times the size of normal. Under such imbalance, plain accuracy is misleading – a classifier can score highly by favouring the majority class – so we report *macro-averaged* F1 and one-vs-rest ROC-AUC as primary metrics, together with per-class precision/recall/F1 and the confusion matrix. Because missing a malignant case is the costliest error, we pay particular attention to *malignant recall*. The imbalance also motivates a *stratified* split and an *inverse-frequency class-weighted* loss (Section 4.1). We expect the malignant and normal classes to be harder: malignant is a minority and visually variable, and normal (no lesion) can be confused

Table 2: BUSI overview (computed in the notebook). "Images with a real lesion" excludes `normal` scans, whose masks are empty.

Property	Value	Property	Value
# images	780	Images with ≥ 1 mask	780
# classes	3	Images with a real lesion	647
Class names	n/b/m	Images with multiple masks	17 (max 3)
Unique image sizes	639	Width range (px)	190–1048
Channels (stored)	RGB (780)	Height range (px)	310–719

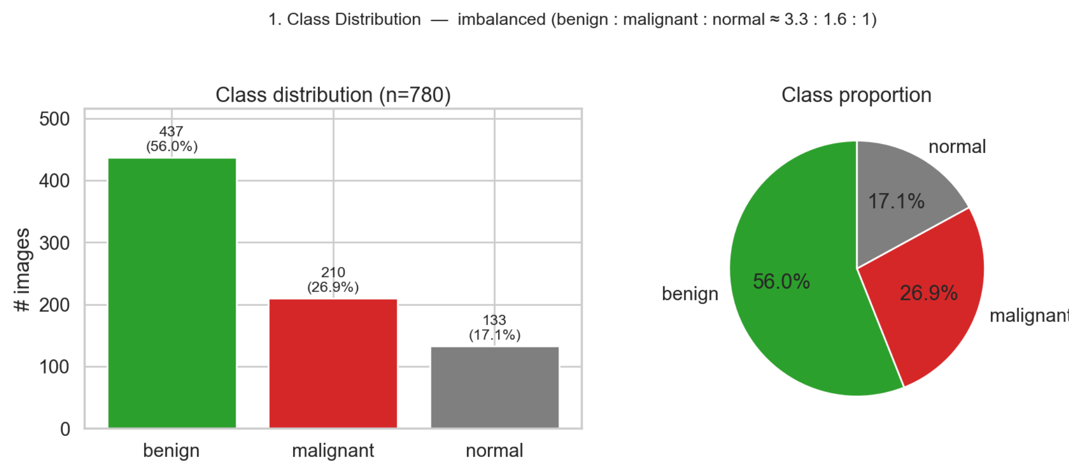


Figure 1: Class distribution of BUSI ($n = 780$). The dataset is imbalanced (benign:malignant:normal $\approx 3.3 : 1.6 : 1$), which drives our use of class weighting and macro-averaged metrics rather than raw accuracy.

with subtle benign findings.

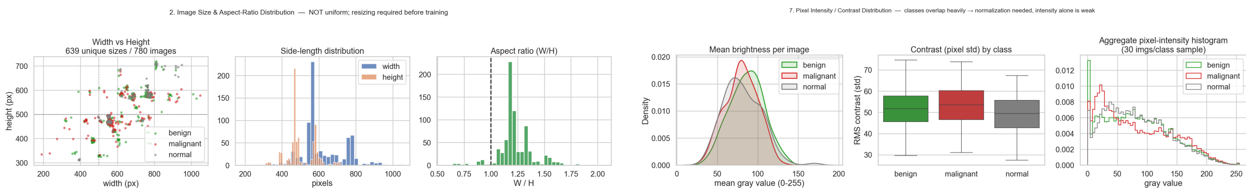
3.3 Image dimensions, channels and intensity

Image sizes vary widely – 639 distinct resolutions, width 190–1048 px and height 310–719 px – and most images are landscape ($W/H > 1$) (Fig. 2a). Although ultrasound is intrinsically grayscale, all files are stored as three-channel *pseudo-RGB* (the channels are essentially identical). Two implications follow: networks require a *fixed* input size, and naive resizing would distort lesion aspect ratio, which is discriminative (Section 3.4); we therefore resize *with padding* (letterboxing) to 256×256. We can safely convert to a single grayscale channel to save computation, or replicate to three channels to reuse ImageNet-pretrained weights – we use whichever the model requires.

Per-image brightness and contrast distributions overlap heavily across the three classes (Fig. 2b), so global intensity is a *weak* discriminator on its own: the model must rely on texture and shape. The wide brightness range across images argues for per-image (or ImageNet) normalisation to stabilise training.

3.4 Lesion size, location and morphology

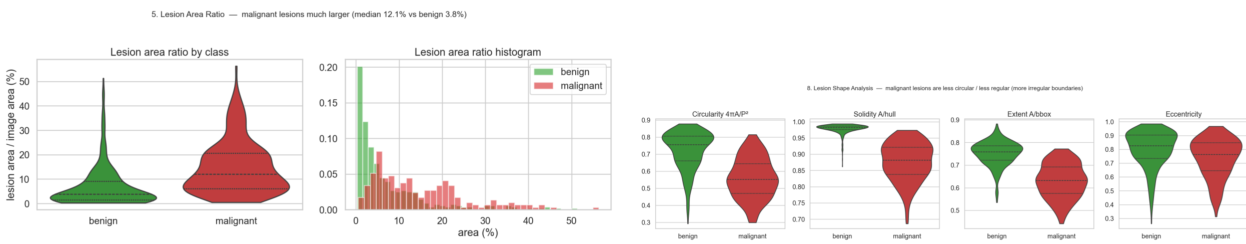
The masks let us quantify lesion geometry. The lesion-area ratio (lesion pixels ÷ image pixels) is much larger for malignant lesions (median $\approx 12\%$) than for benign lesions (median $\approx 4\%$) (Fig. 3a).



(a) Image dimensions / aspect ratio.

(b) Per-class pixel-intensity / contrast.

Figure 2: Image properties. (a) Sizes are inconsistent and mostly landscape, motivating letterbox resizing to a fixed 256×256. (b) Brightness and contrast overlap across classes, so intensity alone is a weak feature and per-image normalisation is used.



(a) Lesion-area ratio by class.

(b) Lesion shape descriptors by class.

Figure 3: Lesion morphology. (a) Malignant lesions are larger (median $\approx 12\%$ vs. $\approx 4\%$ of the image); many benign lesions are tiny. (b) Malignant lesions are less circular/solid and more eccentric, i.e. more irregular. Both findings inform loss choice, model choice and the shape-feature experiment.

This has two consequences: lesion *size is itself discriminative*, and many benign lesions are *very small*, creating a strong pixel-level foreground/background imbalance that favours an overlap-aware segmentation loss (Dice/Tversky rather than pixel accuracy). Aligned per-class mask heatmaps show that lesions concentrate *near the image centre, slightly above the middle* (Fig. 4), an acquisition bias arising because sonographers centre the probe on the finding; random crops/translations during training reduce over-reliance on this position cue. Finally, shape descriptors computed from the masks (circularity, solidity, extent, bounding-box aspect and eccentricity) show that malignant lesions have *lower circularity, solidity and extent and higher eccentricity* than benign lesions (Fig. 3b), quantifying their "irregular, spiculated" appearance. These shape differences later motivate two things: boundary-aware/attention segmentation models, and the shape/radiomics feature-fusion experiment in the dual-stage classifier (Section 4.7).

3.5 Duplicate images, label conflicts and leakage risk

This is the most consequential integrity check. We detect *exact* duplicates via MD5 hashing of file bytes and *near*-duplicates via a perceptual hash (pHash) with Hamming distance. We find an **exact cross-class duplicate**: `benign (433).png` and `malignant (145).png` are the same file bytes but carry *different* class labels – a genuine labelling conflict. More broadly, at a Hamming threshold of 5 there are 186 near-duplicate pairs, of which 10 *cross class boundaries*; these group into 124 clusters covering 277 images (36% of the data) (Fig. 5a). This is consistent with a documented BUSI data-quality issue [28].

Near-duplicates are dangerous for evaluation. Simulating naive random 70/30 image-level splits, on the order of 57 near-duplicate clusters end up with copies on *both* the train and test sides (Fig. 5b); the test set then contains images the model effectively saw in training, which inflates the reported

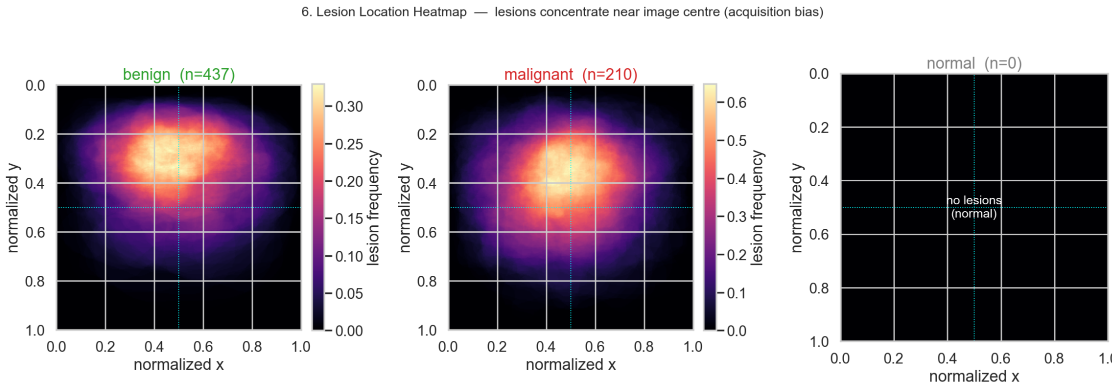
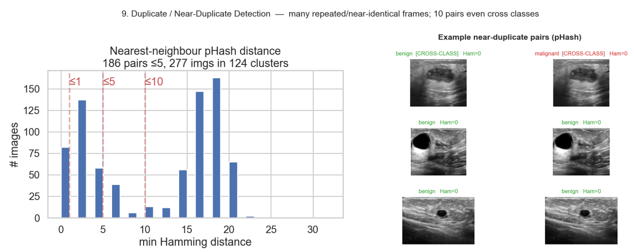
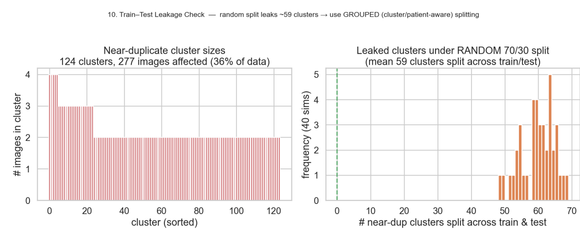


Figure 4: Aligned per-class lesion-location heatmaps. Lesions concentrate near the image centre (an acquisition bias); `normal` has no lesion signal. This motivates translation/crop augmentation so the model does not simply exploit lesion position.



(a) Near-duplicate detection (pHash).



(b) Leakage under random splitting.

Figure 5: Data-quality analysis. (a) 186 near-duplicate pairs (10 cross-class) form 124 clusters covering 36% of BUSI, including an exact cross-class duplicate. (b) Simulated random 70/30 splits leak dozens of clusters across train/test (mean on the order of tens of clusters), motivating a cluster-grouped split. Panel (b) shows one 40-run simulation.

accuracy and Dice. Our remedy is threefold: drop the conflicting exact duplicate, merge multi-masks by union, and split at the *cluster* level so that all near-duplicates of an image stay on a single side, while stratifying by class. Table 3 summarises the cleaning and grouping. This single decision is the main reason our reproductions score below their published values (Section 6).

3.6 Pre-processing decisions

Every pre-processing choice is tied to an EDA observation rather than adopted by default (Table 4). We *resize with padding* (letterbox) to 256×256 to preserve lesion aspect ratio; convert to grayscale for from-scratch models or replicate to three ImageNet-normalised channels for pretrained backbones; and normalise per image. On the training split only we apply light augmentation (horizontal flips, small affine transforms and brightness/contrast jitter), which addresses the small, imbalanced dataset and the lesion-centre bias. Contrast-limited adaptive histogram equalisation (CLAHE) and speckle denoising were considered but *not* adopted in the shared pipeline: BUSI images are low-contrast, but speckle is informative texture and aggressive enhancement or denoising can distort it, so we treat these as optional steps to be justified by an ablation rather than used blindly. We ran no dedicated CLAHE/denoising ablation within the compute budget, so we state this as a deliberate omission rather than as evidence that the two steps are useless; it is listed in future work

Table 3: Cleaning and grouping summary (from the notebook's data-quality analysis).

Step	Value
Exact-duplicate file pairs (MD5)	1 (cross-class conflict)
Near-duplicate pairs (pHash Hamming ≤ 5)	186
of which cross-class	10
Near-duplicate clusters	124
Images inside a cluster	277 (36% of data)
Clusters leaked by a naive random 70/30 split	≈ 57 (simulated)
Mitigation	drop conflict; union masks; cluster-grouped stratified split

Table 4: Pre-processing decisions and their EDA justification.

Step	Decision	Justification (EDA)
Resize to 256×256	required	Inconsistent sizes (§3.3); fixed input needed.
Letterbox (pad, keep ratio)	preferred	Mostly landscape; naive resize distorts discriminative lesion shape (§3.4).
Normalisation (per-image/ImageNet)	required	Wide, overlapping brightness (§3.3); stabilises training.
Grayscale vs. 3-channel	model-dependent	Pseudo-RGB storage; 1 channel saves compute, 3 channels reuse ImageNet weights.
Augmentation (flip/affine/jitter)	training only	Small, imbalanced data + centre bias (§3.2,3.4).
CLAHE / denoising	not adopted	Low contrast but informative speckle; risk of distortion – ablation only.
De-duplicate + cluster split	critical	Duplicates and leakage (§3.5) would inflate metrics.

(Section 7.7).

3.7 EDA implications for model design

Table 5 maps each headline EDA finding to a modelling decision. These decisions carry directly into Sections 4 and 5.

Table 5: From EDA observation to modelling implication.

EDA observation	Modelling implication
Small dataset overall	Transfer learning + augmentation + regularisation.
Class imbalance	Stratified split, class-weighted loss, macro-F1 + AUC.
Many tiny benign lesions	Overlap-aware loss (Dice/Tversky); report Dice/IoU; analyse recall.
Irregular malignant boundaries	Attention / multi-scale segmentation models.
Near-duplicates	Cluster-grouped, leakage-free split.
Benign vs. malignant shape differences	Evaluate shape/radiomics features in the classifier.
Lesion-centre acquisition bias	Random crop/translation augmentation.
Overlapping intensity	Per-image normalisation; rely on texture/shape.

Table 6: Overview of all models. "Origin": R = reproduced published model, P = proposed by us. "Backbone": pretrained (PT) or from scratch (FS). Parameter counts are read from the result files.

Model	Task	Backbone / main idea	PT/FS	Loss	Params (M)	Origin
U-Net [3]	Seg	encoder–decoder + skips	FS	Dice+BCE	31.04	R
Attention U-Net [9]	Seg	U-Net + attention gates	FS	Dice+BCE	31.39	R
FCN-AlexNet [2]	Seg	AlexNet → fully conv.	PT	Dice+BCE	28.70	R
Patch-LeNet [2]	Det	28×28 patch classifier	FS	CE (patches)	0.43	R
DeepLabV3+ [7]	Seg	ResNet-50 + ASPP	PT	BCE+soft-Dice	26.69	R
nnU-Net [20]	Seg	blueprint 2D U-Net + deep sup.	FS	CE+Dice (DS)	8.55	R
AAU-Net [12]	Seg	hybrid adaptive attention	FS	Dice+BCE	42.75	R
CResU-Net [13]	Seg	Co-Block/residual schedule	FS	Dice	33.54	R
ResNet-18 [4]	Cls	residual backbone	PT	weighted CE	11.18	R
DenseNet-121 [5]	Cls	dense connectivity	PT	weighted CE	n/r	R
EfficientNet-B0 [6]	Cls	compound-scaled MBConv	PT	weighted CE	n/r	R
Dual-stage [17]	Both	DeepLabV3+ → ROI → EffB0	PT	weighted CE	–	R
MTL-OCA [25]	Both	shared Res-UNet + OCA + cls	FS	0.4CE$_s$+CE+CE	8.19	R
Improved U-Net (ours)	Seg	U-Net + **PT EffB4** enc.	PT	Focal-Tversky+BCE	20.23	**P**
Improved dual-stage (ours)	Both	**Improved-UNet** → ROI → EffB0	PT	weighted CE	–	**P**

4 Models and Methods

We reproduce thirteen published models and build two of our own. All models share the same cleaned, cluster-grouped, leakage-free split, letterbox pre-processing and metrics from `Models/common/`, so differences in the numbers reflect the architecture and training recipe, not data handling. Table 6 gives an overview; we then describe each model consistently, noting which parts are pretrained and which are our own implementation. For each model we implemented the head/decoder and the training/evaluation loop ourselves; pretrained backbones are reused torchvision or `segmentation-models-pytorch` (smp) components, as indicated.

4.1 Loss functions and metrics

Let $p_i \in [0,1]$ be a predicted (foreground) probability and $g_i \in \{0,1\}$ the ground truth over N pixels (segmentation) or examples (classification).

Binary cross-entropy (BCE).

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\Big[g_i \log p_i + (1-g_i)\log(1-p_i)\Big]. \tag{1}$$

Weighted multi-class cross-entropy. For C classes with softmax output $p_{i,c}$ and one-hot label $g_{i,c}$,

$$\mathcal{L}_{\text{wCE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} w_c\, g_{i,c} \log p_{i,c}, \qquad w_c \propto \frac{N}{C\, n_c}, \tag{2}$$

where n_c is the number of training examples of class c; the *inverse-frequency* weights w_c counter the imbalance of Section 3.2.

Dice coefficient and Dice loss. The Dice similarity coefficient (DSC) and its soft loss are

$$\mathrm{DSC} = \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \qquad \mathcal{L}_{\mathrm{Dice}} = 1 - \mathrm{DSC}, \tag{3}$$

with ϵ a small constant for numerical stability. DSC emphasises overlap and is well suited to the small-lesion foreground/background imbalance of Section 3.4.

Intersection-over-Union (IoU/Jaccard).

$$\mathrm{IoU} = \frac{|P \cap G|}{|P \cup G|} = \frac{\sum_i p_i g_i}{\sum_i p_i + \sum_i g_i - \sum_i p_i g_i}. \tag{4}$$

Tversky index and Focal-Tversky loss. The Tversky index generalises Dice by weighting false positives (FP) and false negatives (FN) asymmetrically:

$$T = \frac{\sum_i p_i g_i}{\sum_i p_i g_i + \alpha \sum_i p_i (1 - g_i) + \beta \sum_i (1 - p_i) g_i}, \tag{5}$$

and the Focal-Tversky loss [26, 27] adds a focusing exponent γ:

$$\mathcal{L}_{\mathrm{FT}} = (1 - T)^{\gamma}. \tag{6}$$

Choosing $\beta > \alpha$ penalises false negatives (missed lesion pixels) more heavily, targeting the missed-lesion tail identified in Section 3.4. Setting $\alpha = \beta = 0.5$, $\gamma = 1$ recovers the Dice loss exactly, so Focal-Tversky is a strict generalisation and the ablation of Section 6.1 is a controlled one.

Our combined segmentation loss. The improved segmenter is trained with a convex blend of BCE and Focal-Tversky,

$$\mathcal{L}_{\mathrm{combo}} = \lambda\, \mathcal{L}_{\mathrm{BCE}} + (1 - \lambda)\, \mathcal{L}_{\mathrm{FT}}, \qquad \lambda = 0.5, \quad \alpha = 0.3, \quad \beta = 0.7, \quad \gamma = \tfrac{4}{3}, \tag{7}$$

with the Tversky smoothing constant $\epsilon = 1$ and the index computed *per image* before averaging over the batch (so a large lesion cannot dominate a small one). The values $(\alpha, \beta, \gamma) = (0.3, 0.7, 4/3)$ are the defaults recommended by Abraham and Khan [27] and were *not* tuned on our data, which keeps the ablation honest. The BCE term keeps pixel-level gradients healthy early in training, when the Tversky term is nearly flat; the Focal-Tversky term then drives the overlap metric and the false-negative penalty. The identical loss (same $\lambda, \alpha, \beta, \gamma$) is used to generate the out-of-fold training masks of Section 4.7.

Classification metrics. With per-class precision P_c and recall R_c, the class F1 and macro-F1 are

$$\mathrm{F1}_c = \frac{2 P_c R_c}{P_c + R_c}, \qquad \text{macro-F1} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{F1}_c. \tag{8}$$

Macro-F1 weights every class equally and is our primary classification metric (Section 5.4); we also report balanced accuracy, accuracy, macro one-vs-rest ROC-AUC and the confusion matrix. Segmentation is scored with *per-image* Dice and IoU (averaged over images, so large lesions do not dominate), pixel precision/recall, and the Yap detection metrics (true-positive fraction, false positives per image, F-measure) where applicable [2].

4.2 Reproduction: the project's three required papers

The brief names three papers – the BUSI dataset [1], Yap et al. [2] and U-Net [3] – so we first reproduce the three models these define.

U-Net (from scratch) [3]. A four-level contracting/expanding encoder–decoder of double 3×3 convolutions with 2×2 pooling and skip connections that concatenate encoder features into the decoder; we pad convolutions to preserve size, add batch normalisation, and output a single-channel logit map from a $1 \times 256 \times 256$ grayscale input, trained with Dice+BCE. *Own work:* the whole architecture and training loop. It is our from-scratch segmentation reference and the baseline of the improved-segmenter ablation.

Patch-based LeNet (detection) [2]. A small LeNet classifies 28×28 grayscale patches as lesion vs. non-lesion; at test time a sliding window builds a lesion-probability map that is thresholded into a detection map. We train it with balanced hard-negative sampling. This is a *detection* method, scored with the Yap detection metrics. *Own work:* the LeNet, the balanced/hard-negative patch sampler and the sliding-window inference.

FCN-AlexNet (Yap's best method) [2]. An ImageNet-pretrained AlexNet is made fully convolutional (fc6/fc7 recast as convolutions, a 1×1 score convolution and bilinear upsampling) and fine-tuned for segmentation. *Own work:* the fully convolutional head and fine-tuning; the AlexNet backbone is a pretrained component.

4.3 Additional segmentation models

Attention U-Net [9] augments our U-Net with an additive attention gate on every skip connection, computing an attention map $\alpha = \sigma(\psi\,\mathrm{ReLU}(W_x x + W_g g))$ and re-weighting the skip as αx. It uses the *same* loss and recipe as our U-Net, so it is a controlled test of the attention gate. *Own work:* the attention-gate module. **DeepLabV3+ [7]** uses an ImageNet-pretrained ResNet-50 encoder (output stride 16) feeding an ASPP head and a light decoder; it is our strongest *reproduced* segmenter. *Own work:* the ASPP head and decoder; the ResNet-50 is pretrained. **nnU-Net [20]** is reproduced as its fixed *blueprint* 2D U-Net (Conv–InstanceNorm–LeakyReLU, strided-conv down, transposed-conv up, deep supervision), trained with the nnU-Net recipe; we deliberately drop the self-configuring framework, so this is an architecture-and-recipe reproduction. *Own work:* the network, deep-supervision loss and loop. **AAU-Net [12]** reproduces the Hybrid Adaptive Attention Module (multi-receptive-field convolutions with channel and spatial self-attention); *own work:* the HAAM module and network, written from scratch. **CResU-Net [13]** schedules Co-Block (dense-concatenation), ResNet-identity and MultiRes encoder blocks and is trained with a pure Dice loss (paper-faithful); *own work:* the three block types and the block-scheduled U-Net.

4.4 Classification models

All three backbones take a three-channel ImageNet-normalised 256×256 input (grayscale replicated to pseudo-RGB), have their final layer replaced by a fresh three-way head, and are fine-tuned with inverse-frequency weighted cross-entropy. We deliberately use the *smallest* variant of each family because BUSI is small. **ResNet-18 [4]** is fine-tuned with SGD; **DenseNet-121 [5]** adds dropout before the head and is trained with early stopping on validation macro-F1; **EfficientNet-B0 [6]** adds dropout before a three-way head. *Own work:* the heads and training pipelines.

4.5 Combined segmentation–classification

Dual-stage pipeline (reproduced) [17]. Stage-1 DeepLabV3+ predicts the lesion mask; from the predicted mask we take the largest connected component, expand its bounding box and crop the ROI (falling back to the full frame if no lesion is found); Stage-2 is an EfficientNet-B0 with a fresh two-way head trained on the ROI crops to classify benign vs. malignant. We follow the paper's two-class setting and reuse our ResNet-50 DeepLabV3+ as Stage-1. *Own work:* the ROI coupling (mask → bounding box → crop) and the end-to-end pipeline. The coupling is deterministic and is worth stating precisely, because it is held *fixed* across every dual-stage variant we compare: on the binarised 256×256 mask we run 8-connected component labelling, keep the component of largest area, take its axis-aligned bounding box (x, y, w, h) and dilate it by a relative margin of 15% per side ($x_0 = \max(0, x - 0.15w)$, $x_1 = \min(W, x + w + 0.15w)$, and likewise in y), clipping to the image. Boxes whose shorter side falls below 16 px are rejected as unusable. If the box is rejected or the mask is empty, the *whole frame* is passed to Stage-2 – the paper's behaviour, so that a missed segmentation still produces a prediction rather than an abstention. The crop is finally resized to 256×256 for Stage-2. We report both the full pipeline score (predicted-mask ROI) and a *ground-truth-ROI (GT-ROI) upper bound*, in which Stage-2 is fed ground-truth-mask crops – a diagnostic for perfect segmentation. **Multi-task MTL-OCA (reproduced) [25].** A shared Res-UNet backbone feeds an Object Contextual Attention segmentation head and a global-average-pooled MLP three-class head, trained with the combined loss $\mathcal{L} = 0.4\,\mathrm{CE}_{\mathrm{soft}} + \mathrm{CE}_{\mathrm{final}} + \mathrm{CE}_{\mathrm{cls}}$. The paper's "OASBUD" statistics match BUSI, so we reproduce it on BUSI. *Own work:* the Res-UNet, OCA head, classification head and multi-task loop.

4.6 Our improved segmenter

Our improvement is framed as a hypothesis rather than a mere architecture: the only reproduced segmenter that used transfer learning (DeepLabV3+) clearly out-scored the from-scratch ones (Section 6), so we hypothesise that *transfer learning is the decisive factor* for BUSI segmentation. We therefore build a U-Net whose **encoder is an ImageNet-pretrained EfficientNet-B4** (via smp), keeping the U-Net decoder and skip structure. On top of this we test two further ideas suggested by the EDA: a **Focal-Tversky + BCE** loss (Eq. 7) to attack the missed-lesion tail (Section 3.4), and **test-time augmentation (TTA)** (horizontal-flip and multi-scale) with **largest-connected-component post-processing**.

The inference path is fully specified as follows, since two of the four ablation rows differ only in it. TTA averages the sigmoid probability maps of the original image, its horizontal flip (un-flipped again before averaging) and two rescaled copies (resized, forwarded, resized back), so the network is queried four times per image; averaging probabilities rather than binary masks avoids a hard vote on uncertain pixels. The averaged map is thresholded at a fixed 0.5, closed with an elliptical 5×5 morphological `CLOSE` to seal small holes along fuzzy boundaries, and reduced to its largest 8-connected component with components below 30 px discarded. The single-lesion assumption behind that last step is justified by the EDA: after union-merging, essentially every image carries one lesion region (Section 3.1). We evaluate each ingredient in a controlled ablation whose components are: (a) the from-scratch U-Net baseline; (b) + pretrained EfficientNet-B4 encoder (loss still Dice+BCE); (c) + Focal-Tversky loss; (d) + TTA+post-processing (the full model). *Own work:* the model assembly, loss, TTA/post-processing and the ablation.

4.7 Our improved dual-stage pipeline

Starting from the reproduced dual-stage pipeline we make three changes and then evaluate them rigorously. *(i) Stronger Stage-1 (controlled substitution).* We replace Stage-1 with our improved segmenter and keep the ROI coupling and the Stage-2 classifier *byte-for-byte identical*, so any change in the end-to-end score is attributable to segmentation alone. *(ii) Strengthened Stage-2.* The original pipeline trains Stage-2 on clean ground-truth-mask ROI crops but tests it on imperfect predicted-mask crops, a train/test distribution mismatch. We instead train Stage-2 on *both* ground-truth and predicted-mask ROI crops, select on predicted-ROI validation macro-F1, and add horizontal-flip TTA. *(iii) Feature fusion.* From the predicted mask we optionally compute *shape* features (6 descriptors) or *radiomics* features (14) and concatenate them to the pooled EfficientNet-B0 embedding before the two-way head, so the classifier sees deep features and explicit geometry together. The six shape descriptors are exactly the EDA descriptors of Section 3.4, computed on the largest connected component of the predicted mask (all zero if the mask is empty):

$$\text{area ratio} = \frac{A}{HW}, \quad \text{circularity} = \frac{4\pi A}{P^2}, \quad \text{solidity} = \frac{A}{A_{\text{hull}}}, \quad \text{extent} = \frac{A}{wh}, \tag{9}$$

plus the bounding-box aspect w/h and the eccentricity of the best-fit ellipse, where A and P are the component's area and perimeter, A_{hull} its convex-hull area and (w, h) its bounding-box size. The 14-D radiomics vector extends these with 3 intensity statistics (mean, standard deviation and median grey level *inside* the mask) and 5 Haralick GLCM texture statistics (contrast, dissimilarity, homogeneity, energy, correlation), computed on the grey levels quantised to 32 bins at offset 1 and averaged over the 0° and 90° directions; these are the classical BUS-CAD features of the Cheng and Xian surveys [18, 19]. Crucially, the features are computed from the *predicted* mask at both training and test time, so the classifier never sees geometry it would not have at deployment.

Crucially, we evaluate these candidate improvements rigorously. To train Stage-2 on predicted-mask crops we need predicted masks for the *training* images; if these came from a segmenter that had already seen those images the masks would be unrealistically clean. We therefore generate them *out-of-fold (OOF)* by five-fold cross-fitting, so each training image's mask comes from a segmenter that never saw it – a correctness measure, not a way to raise the score. Every configuration is run as a *five-seed* ensemble and we report a bootstrap 95% confidence interval on the 98-image test set. *Own work:* the Stage-1 substitution, the strengthened/OOF Stage-2 protocol, the shape/radiomics fusion and the multi-seed+CI evaluation.

On comparing the two headline numbers. The segmenter's Dice (0.760) and the pipeline's macro-F1 (≈ 0.87) measure *different tasks* on different test subsets and are not directly comparable. A mask with Dice 0.76 still localises the lesion well enough to crop a useful ROI; the classifier then operates on the cropped pixels and does not need a pixel-perfect boundary. Indeed the predicted-mask pipeline matches the GT-ROI upper bound (Section 6.3), confirming that segmentation quality is not the bottleneck.

5 Experimental Setup

5.1 Data split and leakage prevention

All models use one cluster-grouped, class-stratified split (Section 3.5), built once by `Models/common/data.py` and cached to `split.csv` / `split_cls.csv` so that every training script reads the *same* file. The procedure is: drop the exact cross-class duplicate pair, union-merge multi-mask images,

Table 7: Composition of the shared leakage-free split (from `Models/common/split.csv` and `split_cls.csv`). "Clusters" counts near-duplicate clusters, the unit that is actually allocated; the last row is the leakage assertion.

	Segmentation (2 classes, 645 imgs)				Classification (3 classes, 778 imgs)				
Split	benign	malig.	total	clusters	normal	benign	malig.	total	clusters
Train	298	147	445	—	95	298	147	540	—
Validation	71	31	102	—	18	71	31	120	—
Test	67	31	98	—	20	67	31	118	—
All	436	209	645	532	133	436	209	778	626
Clusters spanning > 1 split: **0** (segmentation) **0** (classification)									

assign each image a near-duplicate cluster id (pHash, Hamming $\leq$ 5; singletons get their own id), then shuffle the *clusters* of each class with a fixed seed (42) and allocate whole clusters to train/validation/test in 70/15/15 proportion. Because the allocation is done per class, the split is stratified; because it is done on clusters, no near-duplicate can straddle the boundary.

The segmentation task uses the 645 benign+malignant lesion images, split 445/102/98 (train/validation/test) over 532 clusters; the three-class classification task uses all 778 retained images, split 540/120/118 over 626 clusters; the dual-stage pipeline is evaluated on the 98-image two-class (benign/malignant) test subset, which is by construction a subset of the 118-image three-class test set, so the two tasks are scored on consistent data. Table 7 gives the per-class composition. A programmatic sanity check (`data._report_split`) confirms that *zero* clusters span more than one split in either task – the assertion that makes the whole benchmark leakage-free. Note that the class proportions of each split match the global 56/27/17 ratio of Section 3.2 to within about one percentage point, so no split is accidentally easier than another. Augmentation is applied to the training split *only*; validation data is used for model selection (checkpoint and threshold), and the test set is untouched during training and tuning.

5.2 Training configuration

Table 9 lists the training configuration of every model, read from the result files. Settings differ between models because each follows its source paper's recipe where practical; we did not force a single shared recipe, since doing so would misrepresent the reproductions. What *is* shared, and is what makes the comparison fair, is everything upstream of the optimiser: the split, the letterbox pre-processing, the augmentation policy of Table 8, the metric code and the test set.

Shared augmentation policy. Augmentation is implemented once in `Models/common/data.py` with Albumentations and applied to the training split only. Its parameters are deliberately conservative, because BUSI is small enough that aggressive photometric distortion destroys the speckle texture the models rely on (Section 3.6). For segmentation the geometric transforms are applied jointly to the image and its mask so the pair stays registered; for classification the same transforms are applied image-only. Table 8 lists the exact settings and ties each to its EDA motivation. Note that we deliberately do *not* use vertical flips or large rotations: an ultrasound frame has a fixed top-down geometry (probe surface at the top, depth increasing downwards), so a vertical flip would produce an anatomically impossible image.

Table 8: Shared training-time augmentation (`Models/common/data.py`). Applied to the training split only; geometric transforms are applied jointly to image and mask for segmentation.

Transform	Parameters	Motivation
Horizontal flip	$p = 0.5$	Left/right breast is arbitrary; doubles effective data.
Affine: scale	[0.9, 1.1]	Lesion size varies with depth/probe setting (§3.4).
Affine: translation	±5% of side	Breaks the lesion-centre acquisition bias (§3.4).
Affine: rotation	±15°	Small probe-angle variation; larger angles are unphysical.
Brightness/contrast	±0.2, $p = 0.3$	Wide inter-image brightness range (§3.3).
Vertical flip, elastic	*not used*	Would violate the fixed top-down US geometry.

Table 9: Training configuration per model (from the result JSON files). LR = initial learning rate; wCE = inverse-frequency weighted cross-entropy; DS = deep supervision. All inputs are 256×256.

Model	Epochs	Batch	Optimiser (LR)	Scheduler	Loss
U-Net	60	8	Adam (10^{-3})	–	Dice+BCE
Attention U-Net	60	8	Adam (10^{-3})	–	Dice+BCE
FCN-AlexNet	60	8	SGD-m0.9 (10^{-3})	–	Dice+BCE
Patch-LeNet	40	256	RMSprop (10^{-2})	–	CE (patches)
DeepLabV3+	60	8	AdamW$_{wd=10^{-4}}$ (10^{-4})	cosine	BCE+soft-Dice
nnU-Net	200	8	SGD-Nesterov0.99 (10^{-2})	poly$^{0.9}$	CE+Dice (DS)
AAU-Net	80	8	Adam (10^{-3})	cosine	Dice+BCE
CResU-Net	120	8	Adam (10^{-3})	cosine	Dice
ResNet-18	60	16	SGD-m0.9,$wd10^{-4}$ (10^{-3})	–	wCE
DenseNet-121	30	16	AdamW$_{wd=10^{-4}}$ (10^{-4})	early stop	wCE
EfficientNet-B0	20	16	Adam (10^{-4})	–	wCE
MTL-OCA	300	16	Adam (10^{-3})	cosine	0.4CE$_s$+CE+CE
Dual-stage (Bruno)	30	16	Adam (10^{-4})	–	wCE
Improved U-Net	60	8	AdamW (10^{-3})	cosine	Focal-Tversky+BCE
Improved dual-stage	60	16	Adam (10^{-4})	cosine	wCE

5.3 Hyper-parameter and threshold selection

Recipes came from the source papers where practical (for example nnU-Net's SGD-Nesterov and poly learning rate, CResU-Net's pure-Dice loss). We adapted a few settings for resource or dataset reasons: we use the *smallest* classification backbones (ResNet-18, EfficientNet-B0) because BUSI is small, and we reuse our ResNet-50 DeepLabV3+ as the dual-stage Stage-1 rather than the paper's ResNet-34. Segmentation predictions are thresholded at a fixed 0.5; the Patch-LeNet detection threshold was tuned on validation data (selected value 0.8). All model selection (checkpoints, thresholds, feature choices) uses the training/validation splits only; the test set is never used for tuning, avoiding test-set overfitting.

5.4 Evaluation metrics

Metrics are separated by task (defined in Section 4.1). *Segmentation* is scored with per-image Dice and IoU (primary), pixel precision and recall, and – for the detection-style comparison – the Yap true-positive fraction, false positives per image and F-measure. *Classification* is scored with macro-F1 (primary), accuracy, balanced accuracy, per-class precision/recall/F1, macro one-vs-rest ROC-AUC and the confusion matrix. Macro-F1 is preferred over raw accuracy because the classes are imbalanced (Section 3.2): accuracy can be high while the minority malignant class is handled poorly, whereas macro-F1 weights each class equally and exposes such failures.

5.5 Robustness and statistical evaluation

For the final dual-stage pipeline we go beyond single-run reporting. Each configuration is trained with *five random seeds*; we report the per-seed mean and standard deviation and a *five-model ensemble* (averaged class probabilities). We quantify test-set uncertainty with a *bootstrap* 95% *confidence interval*: we draw $B = 2000$ bootstrap resamples of the 98-image test set with replacement (fixed resampling seed 0), recompute macro-F1 on each, and report the 2.5th and 97.5th percentiles of the resulting distribution (the percentile bootstrap). Note that this interval quantifies *test-set* uncertainty – how much the score would move if we had drawn a different sample of 98 patients – while the per-seed standard deviation quantifies *training* uncertainty. The two are different sources of variance and we report both; the test-set term is by far the larger one here (± 0.07 versus ± 0.01), which is the reason a 98-image benchmark cannot resolve small differences no matter how many seeds are run.

Predicted training masks are generated by *out-of-fold* five-fold cross-fitting (Section 4.7): the 445 segmentation training images are partitioned into five folds, a segmenter is trained on four folds with the identical recipe of Eq. 7, and it predicts masks for the held-out fold, so every training mask comes from a network that never saw that image. The five configurations we compare (none/shape/radiomics fusion, with and without ensembling) all consume the *same* cached OOF masks, so the comparison between them isolates the feature vector alone. We treat overlapping confidence intervals as evidence that a difference is *not* statistically resolved, and we make no significance claim that a CI does not support.

5.6 Implementation and reproducibility

All experiments were run on a single NVIDIA GeForce RTX 5060 Laptop GPU under Windows 11, with Python 3.11.5, PyTorch 2.10.0 (CUDA 12.8), `segmentation-models-pytorch` 0.5.0 (pretrained encoders), torchvision (ResNet/DenseNet/EfficientNet/AlexNet backbones), Albumentations 2.0.8, OpenCV 4.12.0, scikit-learn 1.6.1 (metrics and the bootstrap), scikit-image (GLCM features) and NumPy 2.2.5. Every model therefore ran on identical hardware, so the wall-clock times of Table 10 are directly comparable.

The repository is organised so that every reported number is traceable. The shared split, datasets, losses and metrics live in `Models/common/`; each model has its own directory `Models/<name>/` containing a training script, a saved checkpoint and a `README.md`; and each run writes a result file `Models/results/*.json` recording its configuration, split sizes, training time and full test metrics. Every table and figure in this report is generated from those JSON files or from `COMP9444_EDA.ipynb`, and no number is typed in by hand. Random seeds are fixed throughout (the shared split and the single-run models use seed 42; the final pipeline uses seeds 0–4 and the bootstrap uses seed 0). Two sources of residual non-determinism remain and we do not claim bit-exact reproducibility: cuDNN kernel selection is not forced into deterministic mode, and the pretrained weights are downloaded at run time.

The cost table carries its own message. Our best segmenter is also among the *cheapest* to train (7.9 min), fifteen times faster than AAU-Net (117.5 min) and seven times faster than the from-scratch U-Net it replaces (58.5 min) – because a pretrained encoder converges in far fewer epochs rather than because the network is smaller. Total compute for the entire study, including the five-seed ensembles and the out-of-fold cross-fitting, is on the order of six GPU-hours, which is what made the multi-seed protocol of Section 5.5 affordable at all.

Table 10: Training cost on the shared hardware (wall-clock minutes, from the result JSON files). Times exclude data caching, which is shared. The dual-stage rows train Stage-2 only; Stage-1 checkpoints are reused.

Model	min	Model	min	Model	min
AAU-Net	117.5	nnU-Net	27.7	ResNet-18	1.7
U-Net	58.5	Attention U-Net	17.9	FCN-AlexNet	1.6
MTL-OCA	46.4	**Improved U-Net**	**7.9**	DenseNet-121	1.4
CResU-Net	31.0	DeepLabV3+	5.6	Patch-LeNet	1.3
Improved dual-stage (5-seed)	17.5	Improved dual-stage (1 seed)	3.6	EfficientNet-B0	0.8

Table 11: Segmentation on the leakage-free BUSI test set (mean over 98 images), ranked by Dice. "σ" is the per-image Dice standard deviation; Pix. P and Pix. R are pixel precision and recall. "Paper" is the paper-reported BUSI Dice where available to us. Best per column in bold.

Model	Type	Dice	σ	IoU	Pix. P	Pix. R	Params (M)	min	Paper
Improved Pretrained-UNet (ours)	**ours**	**0.760**	0.307	**0.688**	0.765	0.790	20.23	7.9	–
DeepLabV3+ [7]	transfer	0.739	0.290	0.652	0.765	0.753	26.69	**5.6**	–
AAU-Net [12]	reprod.	0.730	**0.284**	0.638	0.739	0.759	42.75	117.5	0.775
nnU-Net [20]	reprod.	0.717	0.299	0.627	0.754	0.719	8.55	27.7	–
CResU-Net [13]	reprod.	0.693	0.305	0.600	0.759	0.697	33.54	31.0	0.829
U-Net [3]	scratch	0.648	0.362	0.570	0.775	0.637	31.04	58.5	–
Attention U-Net [9]	scratch	0.629	0.345	0.540	0.701	0.638	31.39	17.9	–
MTL-OCA [25] (seg)	joint	0.612	0.332	0.515	0.737	0.598	8.19	46.4	–
FCN-AlexNet [2]	transfer	0.582	0.310	0.471	0.648	0.647	28.70	1.6	–
Patch-LeNet [2]	detect.	0.285	0.268	0.198	0.641	0.223	0.43	1.3	–

6 Results

All models are evaluated on the same leakage-free test split: segmentation on the 98 benign+malignant test images, classification on the 118 three-class test images, and the dual-stage pipelines on the 98-image two-class subset. Every number is read from `Models/results/*.json`. We report observations here and develop their implications in Section 7.

6.1 Segmentation results

Table 11 and Fig. 6 rank the segmentation models by Dice. Our improved segmenter is best (Dice **0.760**, IoU 0.688), ahead of the strongest reproduced baseline DeepLabV3+ (0.739) and the two BUS-specific SOTA reproductions (AAU-Net 0.730, CResU-Net 0.693). Three observations follow. First, our reproductions score *below* the papers' reported BUSI numbers (AAU-Net 0.730 vs. 0.775; CResU-Net 0.693 vs. 0.829), consistent with removing near-duplicate leakage that the original cross-validation protocols did not. Second, the general-purpose "strong baseline" nnU-Net (0.717) does *not* dominate on this small set; the transfer-learning models beat it. Third, the controlled pair Attention U-Net (0.629) vs. U-Net (0.648), which share an identical recipe, shows the attention gate does not help here.

The pixel precision/recall columns explain *how* the models differ, not only by how much. Almost every model is precision-dominant (Pix. P > Pix. R): the shared failure mode on BUSI is under-segmentation – pixels the model declines to claim – not spurious over-segmentation. The from-scratch U-Net is the extreme case, with the second-highest precision (0.775) but the third-lowest recall (0.637): what it does mark is usually lesion, but it marks far too little. Our improved model is the only one whose recall exceeds its precision (0.790 vs. 0.765), which is the intended effect of

Table 12: Detection-style evaluation (Yap et al. [2] criteria) of the same predictions, over 98 test lesions. TPF = true-positive fraction (lesion recall at the region level); FP/img = false-positive regions per image.

Model	TP	FN	TPF ↑	FP/img ↓	F-measure ↑
Improved Pretrained-UNet (ours)	85	13	0.867	**0.163**	**0.854**
DeepLabV3+ [7]	94	**4**	**0.959**	0.408	0.810
FCN-AlexNet [2]	82	16	0.837	0.245	0.804
nnU-Net [20]	89	9	0.908	0.418	0.781
U-Net [3]	81	17	0.827	0.296	0.779
CResU-Net [13]	88	10	0.898	0.429	0.772
AAU-Net [12]	90	8	0.918	0.622	0.723
Attention U-Net [9]	89	9	0.908	0.969	0.631
Patch-LeNet [2]	70	28	0.714	0.673	0.598
MTL-OCA [25] (seg)	83	15	0.847	1.296	0.539

the false-negative-weighted loss of Eq. 7 and the clinically preferable direction. Patch-LeNet is a degenerate case: at recall 0.223 it recovers only a fifth of the lesion pixels, because a 28×28 sliding window scored independently cannot produce a coherent region – which is exactly why Yap et al. themselves preferred the fully convolutional model.

Table 12 scores the same predictions as a *detection* problem using Yap's criteria, and reorders the field. A model can have mediocre Dice yet find nearly every lesion: DeepLabV3+ attains the best true-positive fraction (0.959, missing 4 of 98 lesions) while our improved model attains the best F-measure (0.854) by emitting only 0.16 false positives per image, the cleanest output of any model. The contrast with Attention U-Net is instructive – it detects almost as many lesions as DeepLabV3+ (TPF 0.908) but scatters 0.97 false-positive regions per image, six times as many as our model, which is what collapses its F-measure to 0.631 and its Dice to 0.629. In a screening setting these two failure modes have very different costs: missed lesions are dangerous, whereas false-positive regions cost radiologist time. We therefore report both views rather than ranking on Dice alone.

Ablation of the improved segmenter. Table 13 isolates the three ingredients. The result is unambiguous: the ImageNet-pretrained encoder is responsible for essentially the *entire* gain, raising Dice from 0.648 to 0.765 (+0.117) and reducing the Dice standard deviation from 0.362 to 0.285 – i.e. far fewer completely-missed lesions. Adding the Focal-Tversky loss does *not* raise Dice (a marginal −0.005, from 0.765 to 0.760); consistent with its design it raises pixel recall from 0.763 to 0.800 at some cost to precision, shifting the operating point toward not missing lesions rather than increasing overlap. TTA and post-processing give no Dice gain on this data. We therefore explicitly *do not* claim that Focal-Tversky improves Dice; its measured effect is on recall.

Two details of Table 13 are worth reading carefully, because they are easy to over-claim. First, the Focal-Tversky row is a genuine *trade*, not a wash: recall rises by 3.7 points (0.763 → 0.800) while precision falls by 2.4 (0.799 → 0.775), and at the region level it recovers two further lesions (86 → 88 true positives out of 98). Dice, being symmetric in FP and FN, is almost blind to this reallocation – which is precisely why we report the pixel-level breakdown rather than resting on the headline number. Second, the TTA row leaves Dice unchanged but cuts false positives by a quarter (0.225 → 0.163 per image) and lifts the detection F-measure from 0.846 to 0.854: the largest-connected-component step deletes small spurious regions that contributed too few pixels to register in Dice. Its Dice σ nevertheless *rises* (0.282 → 0.307), because on images where the largest

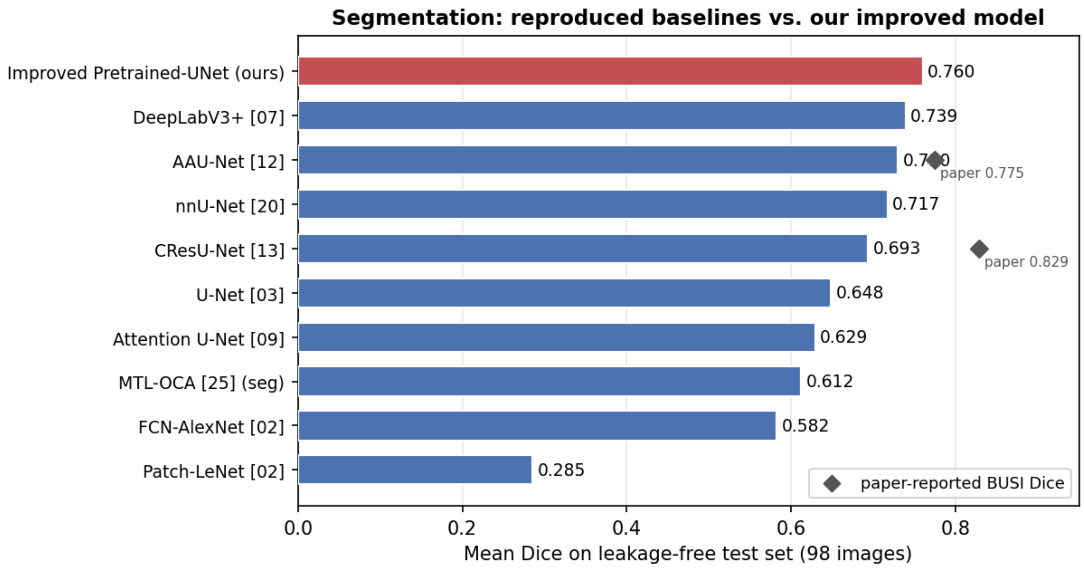


Figure 6: Segmentation Dice on the leakage-free test set. Our improved model (red) leads; the two BUS-specific SOTA reproductions (AAU-Net, CResU-Net) fall below their paper-reported Dice (diamonds) once near-duplicate leakage is removed.

Table 13: Controlled ablation of the improved segmenter (test set, 98 images). Each row adds one component to the previous; everything else – split, pre-processing, augmentation, 60 epochs, AdamW with cosine schedule – is held fixed. Δ is the change in Dice relative to the row above.

Configuration	Dice	Δ	Dice σ	IoU	Pix. P	Pix. R	FP/img
From-scratch U-Net (Dice+BCE)	0.648	—	0.362	0.570	0.775	0.637	0.296
+ ImageNet-pretrained EffB4 encoder	**0.765**	+0.117	0.285	0.685	**0.799**	0.763	0.214
+ Focal-Tversky loss	0.760	−0.005	**0.282**	0.677	0.775	**0.800**	0.225
+ TTA + post-processing (full model)	0.760	±0.000	0.307	**0.688**	0.765	0.790	**0.163**

component is the wrong one, discarding the rest converts a partial hit into a near-total miss. We therefore keep TTA in the final model for its cleaner output while stating explicitly that it does not improve overlap.

Qualitative and failure analysis. Figure 7 shows representative predictions of the improved model: compact, well-localised lesions are segmented accurately, but characteristic failures remain – small low-contrast lesions are partially missed (reducing recall), fuzzy malignant boundaries are over- or under-segmented, and occasional false-positive regions appear in shadowed tissue. The large Dice standard deviation (0.307 for the full model; Table 13) reflects a sub-population of near-completely-missed lesions rather than a uniform small error, which is why the pretrained encoder – which most reduces that tail – matters more than boundary-level refinements.

6.2 Classification results

Table 14 ranks the three-class classifiers by macro-F1. EfficientNet-B0 is best (macro-F1 **0.828**, AUC 0.969), ahead of ResNet-18 (0.810) and DenseNet-121 (0.793); all three pretrained backbones clearly beat the multi-task MTL-OCA classifier (0.720). The confusion matrix of the best model

Table 14: Three-class classification on the leakage-free BUSI test set (118 images: 20 normal, 67 benign, 31 malignant), ranked by macro-F1. Bal. acc. = balanced accuracy; AUC is macro one-vs-rest.

Model	Type	macro-F1	Bal. acc.	Accuracy	AUC	min
EfficientNet-B0 [6]	transfer	**0.828**	**0.852**	**0.848**	**0.969**	**0.8**
ResNet-18 [4]	transfer	0.810	0.820	0.822	0.966	1.7
DenseNet-121 [5]	transfer	0.793	0.804	0.805	0.947	1.4
MTL-OCA [25] (cls)	joint	0.720	0.699	0.746	0.900	46.4

Table 15: Per-class precision (P), recall (R) and F1 on the three-class test set. Support: normal 20, benign 67, malignant 31. Malignant recall, the clinically critical quantity, is highlighted in the last column.

	normal			benign			malignant			
Model	P	R	F1	P	R	F1	P	R	F1	malig. R
EfficientNet-B0	0.667	**0.900**	0.766	**0.919**	**0.851**	**0.884**	**0.862**	0.807	**0.833**	0.807
ResNet-18	0.800	0.800	0.800	0.887	0.821	0.853	0.722	**0.839**	0.776	**0.839**
DenseNet-121	0.773	0.850	**0.810**	0.859	0.821	0.840	0.719	0.742	0.730	0.742
MTL-OCA (cls)	**0.857**	0.600	0.706	0.775	0.821	0.797	0.636	0.677	0.656	0.677

(Fig. 8) shows the expected pattern: the minority *malignant* class is hardest and is the main source of error, and the residual confusion is between benign and malignant (visually similar masses) and between normal and benign. For EfficientNet-B0 the per-class F1 scores are normal 0.766, benign 0.884 and malignant 0.833, with malignant recall 0.807; this per-class view, not the overall accuracy (0.848), is the operative summary for an imbalanced clinical task. MTL-OCA underperforms on *both* tasks (segmentation 0.612, classification 0.720), indicating that sharing one backbone across the two tasks is, on this small dataset, worse than a specialised network per task.

Table 15 breaks the same runs down per class and shows that the macro-F1 ranking hides genuinely different behaviour. *Benign*, the majority class, is easy for all three pretrained backbones (F1 0.84–0.88). *Malignant* is where they separate: EfficientNet-B0 is the most *precise* on malignant (0.862) but ResNet-18 is the most *sensitive* ($R = 0.839$ vs. 0.807), catching one more of the 31 malignant cases. Given that a missed malignancy is the costliest error, ResNet-18 is arguably the better operating point despite ranking second on macro-F1 – a distinction that macro-F1 alone cannot express, and one that a deployment would resolve by tuning the decision threshold rather than by swapping architectures. *Normal* is the noisiest class simply because it has 20 test images, so each one is worth 5 percentage points of recall; EfficientNet-B0's high normal recall (0.900) comes with the lowest normal precision (0.667), i.e. it over-predicts "normal" and absorbs 9 lesion images into it. MTL-OCA fails differently and more seriously: its normal recall is only 0.600, and 11 of 67 benign images are called malignant, so sharing an encoder with the segmentation head costs it most on exactly the distinction that matters clinically.

We did not generate Grad-CAM explanation maps for these classifiers in the current result set, so we make no claim about where the classifier attends; producing and auditing such maps is listed as future work (Section 7.7).

6.3 Dual-stage results

We separate exploratory single-run results (Table 16a) from the rigorous multi-seed results (Table 16b, Fig. 9a).

Controlled Stage-1 substitution. Replacing Bruno's DeepLabV3+ Stage-1 with our stronger segmenter raises Stage-1 Dice from 0.739 to 0.760 (+0.021) but moves the single-run end-to-end macro-F1 only from 0.804 to 0.807 (+0.003). Because the ROI coupling and the Stage-2 network, data and schedule are byte-for-byte identical, this cleanly isolates the segmentation effect, and the transfer ratio is about 0.14 points of F1 per point of Dice – an order of magnitude too small to be worth optimising further.

The GT-ROI diagnostic explains why. Feeding the *same* Stage-2 classifier perfect ground-truth-mask crops lifts macro-F1 to 0.865–0.884, so the "Gap" column of Table 16a – the score still recoverable by making segmentation perfect – is only 0.06–0.08, and it does *not* shrink when Stage-1 improves (Bruno 0.061, ours 0.077). Even a hypothetical Dice-1.0 segmenter would leave the pipeline around 0.88, which is within the confidence interval of what we already achieve by ensembling alone (Table 16b). Put concretely, for the strengthened-Stage-2 row: the pipeline misclassifies 16 of the 98 test images and the GT-ROI oracle misclassifies 10, so handing the classifier perfect masks repairs six errors and leaves ten standing. The residual error is therefore the classifier's own, on lesions that were correctly localised – the system is *classifier-bound*, not segmentation-bound (RQ3).

Rigorous evaluation. The single-run ladder in Table 16a looks like a tidy climb (0.804 → 0.840), but under out-of-fold training masks, five-seed ensembling and bootstrap 95% CIs (Table 16b) the three feature variants' intervals overlap almost entirely (Fig. 9a). Hand-crafted shape/texture features give *no reliable* gain: the ensembles of *none* and *shape* both reach 0.875, and the per-seed means differ by about one standard deviation. The only component that survives is *ensembling* (per-seed 0.847 → ensemble 0.875). On a 98-image test set, single-run differences up to about ±0.05 are not statistically resolved (RQ4). Our best defensible dual-stage result is macro-F1 **0.875** (95% CI $[0.80, 0.94]$).

Panel (c) makes the argument concrete. Re-running the *identical* configuration with only the seed changed moves macro-F1 by up to 0.039 – comparable to the entire 0.036 "improvement" of the single-run ladder in panel (a). The shape variant's own seeds span 0.834 to 0.873, so a study that happened to report seed 2 and compare it against a no-feature seed 1 would "show" a +0.043 gain from shape features that is pure seed noise. This is the mechanism by which single-run ablations on small medical datasets manufacture findings, and it is why we treat panel (a) as exploratory only. Ensembling behaves as variance reduction should: for the two stronger configurations it lifts the score *above* the best individual seed (none: 0.857 → 0.875; shape: 0.873 → 0.875), i.e. the ensemble is better than picking the luckiest run, not merely better than the average one. For radiomics it lands at 0.863, below that configuration's best seed (0.886) but above its mean (0.862) – the expected behaviour when one seed is an outlier rather than genuinely better, and a further reminder that a maximum over seeds is not a result.

6.4 Error analysis

Across tasks the errors share a structure. In classification, the confusion is concentrated on the malignant class and the benign↔malignant boundary (Fig. 8), which matches the EDA finding that benign and malignant lesions overlap in intensity and can resemble one another (Sections 3.3, 3.4); the small normal class is also occasionally confused with benign. In segmentation, errors correlate with lesion difficulty: small, low-contrast lesions and irregular malignant boundaries drive the large Dice standard deviation, and the worst cases are near-complete misses rather than uniform

Table 16: Dual-stage pipelines on the two-class (98-image) test set (67 benign, 31 malignant). (a) single-run ladder (one seed each; indicative only). (b) rigorous evaluation with out-of-fold training masks, five-seed ensembling and a bootstrap 95% CI. (c) the individual seeds behind (b). "GT-ROI" is the perfect-segmentation upper bound (the same Stage-2 fed ground-truth-mask crops); "Gap" is GT-ROI F1 minus pipeline F1, i.e. how much score is still lost to imperfect segmentation. "Malig. R" is malignant recall.

(a) single-run ladder — differences are within test-set noise

Pipeline	Stage-1 Dice	Pipeline F1	Accuracy	AUC	Malig. R	GT-ROI F1	Gap
Bruno (DeepLabV3+ → EffB0) [17]	0.739	0.804	0.827	0.906	0.774	0.865	0.061
Ours: Improved-UNet Stage-1	0.760	0.807	0.827	0.896	0.807	0.884	0.077
Ours: + strengthened Stage-2	0.760	0.817	0.837	0.905	0.807	0.882	0.065
Ours: + shape features	0.761	0.840	0.857	0.936	0.839	0.850	0.010

(b) rigorous: OOF cross-fitting + 5-seed ensemble + bootstrap 95% CI ($B = 2000$)

Pipeline (rigorous)	Ensemble F1	Per-seed mean±std	95% CI	Malig. R	GT-ROI F1
OOF + no hand features (pure CNN)	**0.875**	0.847 ± 0.011	[0.800, 0.936]	**0.903**	0.886
OOF + shape (6 feats)	**0.875**	0.861 ± 0.016	[0.800, 0.936]	**0.903**	**0.906**
OOF + radiomics (14 feats)	0.863	0.862 ± 0.013	[0.785, 0.929]	0.871	0.875

(c) individual seed scores behind panel (b) — the spread that the single-run ladder hides

Features	seed 0	seed 1	seed 2	seed 3	seed 4	range
none	0.857	0.830	0.838	0.855	0.855	0.027
shape	0.850	0.834	0.873	0.873	0.873	0.039
radiomics	0.853	0.855	0.863	0.853	0.886	0.033

boundary error (Fig. 7). This is precisely the sub-population the pretrained encoder most improves (Table 13). In the dual-stage pipeline, because the GT-ROI upper bound is close to the achieved score, the remaining errors are classification errors on correctly localised lesions rather than localisation failures. We report these patterns from the available predictions and figures and do not select only favourable cases.

6.5 Summary of research-question answers

RQ1: under one leakage-free protocol, transfer-learned models lead each task (segmentation: our improved U-Net 0.760, DeepLabV3+ 0.739; classification: EfficientNet-B0 0.828); reproduced BUS-specific SOTA models score below their reported numbers (AAU-Net 0.730 vs. 0.775; CResU-Net 0.693 vs. 0.829). **RQ2:** the pretrained encoder is decisive for segmentation (+0.117 Dice), while loss choice, attention gates, TTA and even nnU-Net add little. **RQ3:** improving segmentation does *not* reliably improve the downstream pipeline; it is classifier-bound (GT-ROI upper bound sits far above the achieved score). **RQ4:** apparent single-run gains are largely not stable – only ensembling survives multi-seed and bootstrap evaluation (best macro-F1 0.875, 95% CI $[0.80, 0.94]$).

7 Discussion

7.1 Effect of evaluation hygiene

The single most important factor shaping our numbers is evaluation hygiene. BUSI contains roughly one third near-duplicate images and an exact cross-class duplicate; a naive random split leaks visually identical frames across train and test and inflates scores. Under our cluster-grouped, leakage-free split, reproduced SOTA models fall well below their published values (CResU-Net 0.829 → 0.693

Dice; AAU-Net 0.775 $\rightarrow$ 0.730). This does not mean our implementations are weak; it means the published numbers were obtained without removing near-duplicates. Our results are lower but trustworthy, and comparisons across papers that use different splits and pre-processing should be treated cautiously. The practical implication is that BUSI leaderboard numbers are not directly comparable unless the splitting and de-duplication procedures match.

7.2 Why transfer learning helped

The controlled ablation (Table 13) shows that an ImageNet-pretrained encoder alone lifts Dice from 0.648 to 0.765 and sharply reduces completely-missed lesions (std 0.362 $\rightarrow$ 0.285). The most plausible explanation is that with only 445 training images the network cannot learn robust low-level visual features from scratch; a pretrained encoder supplies them and stabilises optimisation, which also explains the much shorter training time (58.5 $\rightarrow$ 7.9 min). By contrast, a bespoke Focal-Tversky loss, test-time augmentation, the attention gate of Attention U-Net and the self-configuring nnU-Net add little *overlap* here. We are careful not to over-generalise: attention is not ineffective in general – it simply did not improve performance reliably in this project's small, noisy setting, where the dominant error is missed lesions rather than imperfect boundaries. The Focal-Tversky loss is likewise not useless – it trades precision for higher lesion recall (0.763 $\rightarrow$ 0.800), a clinically sensible operating point – but it does not raise Dice.

7.3 Why the dual-stage pipeline was classifier-bound

A much stronger Stage-1 barely moved the end-to-end score, and the perfect-segmentation GT-ROI upper bound ($\approx$ 0.87–0.88) sits far above the achieved single-run score. Two things follow. First, a rough mask (Dice 0.76) is already good enough to crop a useful ROI, so segmentation is not the bottleneck. Second, the residual error lives in Stage-2, so effort should go there. Our strengthened Stage-2 – trained on both ground-truth and predicted ROI crops to fix the clean/imperfect distribution mismatch – and, above all, ensembling are where the gains came from. This is a transferable lesson about allocating model-development effort: measure the bottleneck (here with a GT upper bound) before investing in the more visible component.

7.4 Statistical uncertainty

The test set has only 98 images, so single-seed rankings are unstable and small differences are within noise. The bootstrap 95% CIs are wide ($\approx [0.80, 0.94]$) and overlap across the feature variants, which is exactly why we decline to claim that shape or radiomics features help. Reporting per-seed mean$\pm$std, an ensemble and a CI turns an over-confident single number into an honest one, and it is the reason our headline dual-stage result is stated with its interval. This is the project's central methodological contribution.

7.5 Clinical and practical interpretation

We interpret the system cautiously as *decision support*, not autonomous diagnosis. Because missing a malignancy is the costliest error, malignant recall matters more than overall accuracy; the segmentation mask and ROI additionally provide localisation and a degree of interpretability that a bare label lacks. There is an inherent sensitivity/false-positive trade-off (the Focal-Tversky operating point is one example), and the appropriate balance is a clinical, not purely statistical, choice. None of the reported numbers, and none of the wide-CI comparisons, would justify unsupervised clinical use.

7.6 Limitations and threats to validity

Several limitations qualify our conclusions. (i) All results come from a single small public dataset with one leakage-free split; even with OOF/multi-seed/CIs the test set cannot resolve small differences. (ii) There is no external, multi-centre validation. (iii) BUSI has near-duplicate and possible annotation issues (including the cross-class conflict we removed). (iv) We use static 2D frames, not full ultrasound sequences, and the data carry acquisition/centre bias (lesion-centre placement). (v) Results are sensitive to pre-processing and thresholds, and our hyper-parameter search was limited by compute. (vi) Our reproductions may differ in detail from the original implementations (for example loss or backbone substitutions, noted per model), so the reproduced numbers characterise *our* faithful re-implementations under a common protocol rather than the original code. (vii) The dual-stage classification is two-class (benign/malignant) and is not directly comparable to the three-class classifiers.

7.7 Future work

The limitations point to concrete next steps: external and multi-centre validation; full nested cross-validation under the same leakage-free protocol to remove any remaining selection effect; uncertainty calibration and Grad-CAM auditing to check that classifiers attend to the lesion rather than to artefacts; stronger or ensembled ROI classifiers, since the pipeline is classifier-bound; ensemble or self-supervised pretraining; adaptation of foundation-model segmenters such as MedSAM under the same protocol; domain-generalisation across scanners; evaluation on full ultrasound video rather than single frames; and a clinician-in-the-loop assessment of the decision-support setting.

8 Conclusion

This project asked how established models perform on breast-ultrasound classification and segmentation under a fair, leakage-free protocol; which design choices matter; whether better segmentation improves downstream classification; and whether apparent gains are statistically stable. To answer these, we detected BUSI's near-duplicate images and built a cluster-grouped, leakage-free split, then reproduced thirteen published models and added two of our own under this one protocol. The evidence is consistent. Removing near-duplicate leakage lowers published BUSI scores substantially (for example CResU-Net Dice $0.829 \rightarrow 0.693$), so our numbers are lower but trustworthy (RQ1). For segmentation, transfer learning is decisive: an ImageNet-pretrained encoder alone lifts Dice from 0.648 to 0.765 and most reduces missed lesions, whereas a bespoke loss, attention gates, test-time augmentation and nnU-Net add little; our improved pretrained U-Net is the best segmenter overall (Dice 0.760) (RQ2). For the combined task, a controlled substitution and a ground-truth-ROI upper bound show the pipeline is bounded by its classifier, not its segmenter (RQ3); and when candidate classifier improvements are evaluated with out-of-fold training masks, five-seed ensembling and bootstrap 95% confidence intervals, the tidy single-run gains largely dissolve – hand-crafted shape/texture features give no reliable improvement and only ensembling survives, giving a best, defensible dual-stage macro-F1 of 0.875 (95% CI $[0.80, 0.94]$) (RQ4). The central methodological contribution is thus a discipline of honest evaluation on small medical datasets: remove leakage, attribute gains with controlled ablations, and report multi-seed results with confidence intervals. The principal limitation is that all findings rest on one small dataset with a wide-interval test set, so external and multi-centre validation is the essential next step. We present this as an experimental decision-support study; it is not suitable for autonomous clinical diagnosis.

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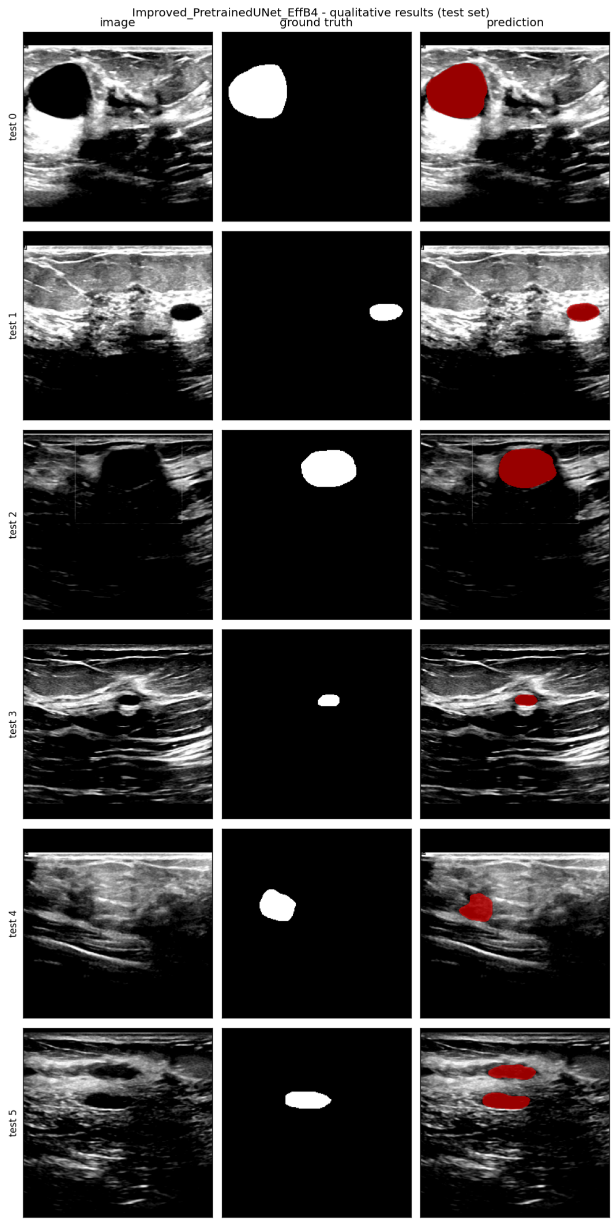


Figure 7: Improved segmenter on test images (input, ground-truth mask, predicted overlay). Compact lesions are captured well; small low-contrast lesions and fuzzy malignant boundaries are the main failure modes.

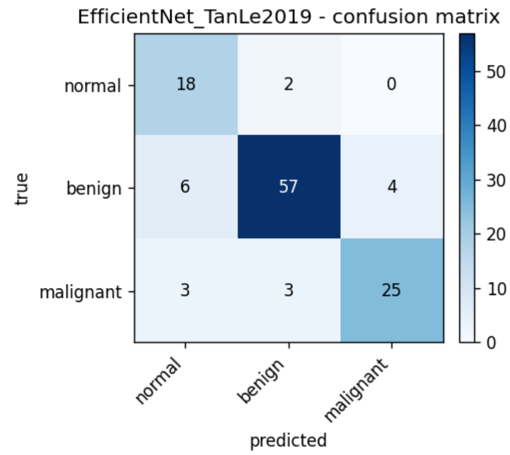
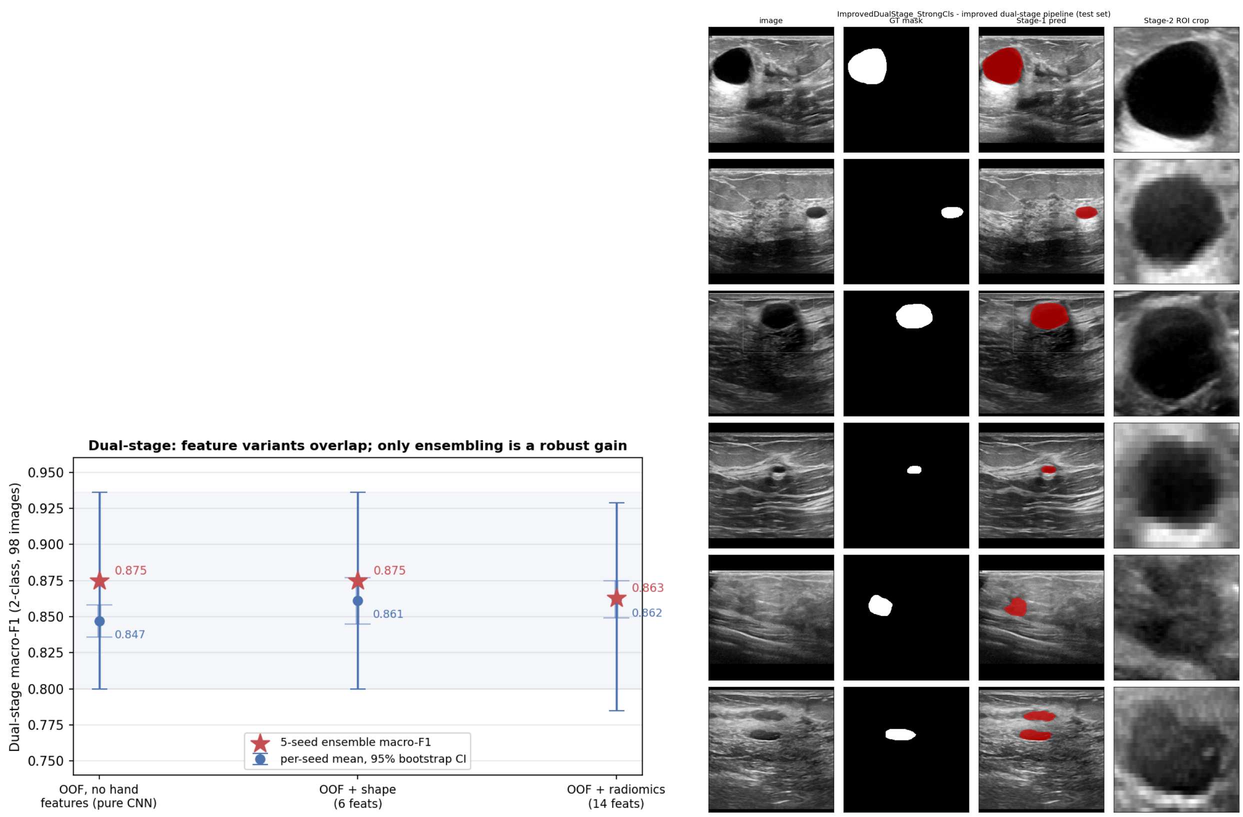


Figure 8: Confusion matrix of the best classifier (EfficientNet-B0, three-class test set). Most error involves the minority malignant class and the benign↔malignant boundary.



(a) Feature variants: per-seed CI vs. ensemble.

(b) Pipeline on test images.

Figure 9: Improved dual-stage pipeline. (a) The three feature variants' 95% confidence intervals overlap almost entirely; only ensembling (stars) reliably lifts the per-seed mean (dots). (b) Qualitative pipeline: input, ground-truth mask, Stage-1 predicted mask, and the ROI crop passed to Stage-2.